Rounding decimals

In this sheet you will...

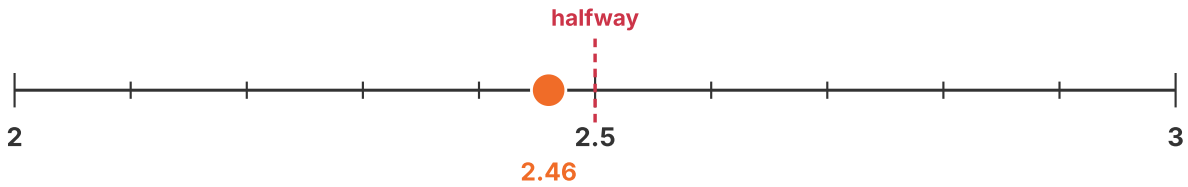
- round a number to a given number of decimal places
- keep a zero at the end when the rounding asks for it
- handle a carry, like 9.96 → 10.0
- avoid the double-rounding trap

Key words

decimal place
round up / down
deciding digit
carry

1 The trap from the starter

Priya rounds **2.46** to the nearest whole number. She says: "The 6 rounds the 4 up to 5, and then the 5 rounds the 2 up to 3. So the answer is 3." Is she right?



No. 2.46 is before the halfway point 2.5, so it is closer to **2**. Priya rounded twice. We must round **once**, looking only at the digit right after the one we keep.

The rounding rule

1. Mark the **last digit you keep**.
2. Look **only** at the next digit, the deciding digit.
3. **5 or more** rounds the kept digit up. **4 or less** leaves it.
4. Cut off everything after it.

Worked examples (round to 1 decimal place)

a **3.46** Keep the 4. The deciding digit is 6, so round up: **3.5**

b **12.04** Keep the 0. The deciding digit is 4, so it stays: **12.0**. The zero stays too, because we were asked for one decimal place.

c **9.96** Keep the 9. The deciding digit is 6, so the 9 becomes 10, and the carry changes the digit before it: **10.0**

Watch out!

Double rounding. 6.249 to 1 d.p. is **6.2**, not 6.3. Only the 4 decides; the 9 after it is cut off.
Zeros at the end stay. 12.0 and 1.00 are right. Writing 12 or 1 does not show the decimal places we were asked for.

Summary checklist

- [ ] I round once, using only the deciding digit.
- [ ] I keep zeros at the end when the rounding asks for them.
- [ ] I know a carry can change the digit before it.

Practice Rounding decimals exercise sheet (p.53)